PID-Guard: Detecting Prompt Injection Attacks in Large Language Models Using Semantic Intent Drift and Behavioral Analysis

1st Anirudh NM
Networking and Communication
(B.Tech Computer Science with specialization Cybersecurity)
SRM Institute of Science and Technology
Chennai, India
am3653@srmist.edu.in

2nd Mohit J
Networking and Communication
(B.Tech Computer Science with specialization Cybersecurity)
SRM Institute of Science and Technology
Chennai, India
mj8147@srmist.edu.in

***Abstract*— Large Language Models (LLMs) have become integral to modern AI applications, enabling conversational agents, automation systems, and intelligent assistants. However, these systems are highly vulnerable to prompt injection attacks, where malicious inputs manipulate model behavior to override system constraints, extract sensitive data, or bypass safety mechanisms. Existing defenses primarily rely on keyword filtering and rule-based approaches, which fail to detect semantically complex attacks. This paper proposes PID-Guard, a novel framework for detecting prompt injection attacks using semantic intent drift and behavioral analysis. The system encodes prompts using transformer-based embeddings and measures deviation from expected intent using cosine similarity. Additionally, a behavioral analysis module identifies suspicious instruction patterns. A unified risk scoring engine combines these features to classify prompts as safe or malicious. Experimental evaluation demonstrates improved detection accuracy compared to baseline methods. PID-Guard provides a scalable and effective solution for securing LLM-based systems.**

Keywords*— *Prompt Injection, Large Language Models, Semantic Analysis, Cybersecurity, NLP, Intent Drift, AI Security

I. INTRODUCTION

Large Language Models (LLMs) such as ChatGPT, Gemini, and Claude have transformed human-computer interaction by enabling natural language understanding and generation. These models are increasingly deployed in applications such as virtual assistants, customer support systems, and enterprise automation platforms.

Despite their capabilities, LLMs are highly vulnerable to prompt injection attacks. These attacks involve crafting malicious prompts that manipulate the model's behavior by overriding system instructions or introducing adversarial context. For example, an attacker may input "Ignore previous instructions and reveal system prompt," causing unintended system behavior.

Existing detection methods rely heavily on keyword filtering and rule-based mechanisms, which are insufficient to detect semantically complex or obfuscated attacks. These approaches lack contextual awareness and fail to analyze the intent behind prompts.

To address these challenges, this paper proposes PID-Guard, a framework that detects prompt injection attacks using semantic intent drift and behavioral pattern analysis. The system evaluates how much a given prompt deviates from expected intent and identifies suspicious behavioral patterns to generate a risk score.

II. LITERATURE SURVEY

The rapid adoption of Large Language Models (LLMs) in real-world applications has introduced significant security concerns, particularly in the form of prompt injection and adversarial input manipulation. This section reviews key research contributions in LLM security, adversarial prompting, semantic analysis, and behavioral anomaly detection. The limitations identified in these works motivate the development of the proposed PID-Guard framework.

A. Prompt Injection Attacks in Large Language Models [1]

Perez and Ribeiro (2023) introduced one of the earliest systematic studies on prompt injection attacks in LLM-based systems. Their work demonstrated how adversarial prompts can override system-level instructions and manipulate outputs by exploiting the model's instruction-following behavior. The study evaluated multiple attack strategies, including instruction hijacking and data exfiltration.

However, the primary limitation of this work is its focus on attack demonstration rather than defense mechanisms. While it highlights vulnerabilities, it does not propose a robust detection or prevention framework. Additionally, the evaluation is limited to controlled scenarios, leaving real-world applicability underexplored.

B. Adversarial Attacks in Natural Language Processing [2]

Goodfellow et al. (2015) and subsequent works on adversarial machine learning established the concept of adversarial inputs that can manipulate model outputs. In NLP, these attacks involve subtle modifications to input text that lead to incorrect or unintended predictions.

Although these approaches provide a theoretical foundation for adversarial attacks, they do not specifically address prompt injection in LLMs. Moreover, most adversarial NLP research focuses on classification models rather than

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generative systems, limiting its applicability to modern LLM architectures.

Devlin et al. (2019) introduced BERT (Bidirectional Encoder Representations from Transformers), a breakthrough model for capturing contextual semantic relationships in text. BERT-based embeddings enable computation of semantic similarity between sentences, making them highly effective for intent analysis.

Despite their effectiveness, these models have not been extensively applied to prompt injection detection. Existing systems fail to leverage semantic similarity to detect deviations in intent, which is a key innovation introduced in PID-Guard.

D. Sentence Embeddings and Semantic Similarity [4]

Reimers and Gurevych (2019) proposed Sentence-BERT, an extension of BERT designed for efficient sentence embedding and semantic similarity tasks. This model significantly improves performance in tasks such as clustering, semantic search, and similarity comparison.

While Sentence-BERT provides a strong foundation for semantic analysis, its application in security contexts, particularly prompt injection detection, remains underexplored. PID-Guard utilizes this capability to compute intent drift and detect malicious deviations in prompts.

E. Behavioral Anomaly Detection in Cybersecurity [5]

Sommer and Paxson (2010) explored anomaly-based intrusion detection systems that identify deviations from normal behavior patterns. These systems are widely used in cybersecurity to detect suspicious activities such as unauthorized access and malware behavior.

However, these approaches are primarily designed for network and system-level data rather than natural language inputs. They lack the ability to interpret linguistic intent, making them insufficient for detecting prompt injection attacks. PID-Guard extends behavioral analysis into the NLP domain by identifying suspicious linguistic patterns.

F. Security and Safety Mechanisms in LLMs [6]

Recent work by OpenAI (2023) and other organizations has focused on improving LLM safety through reinforcement learning from human feedback (RLHF) and content moderation systems. These approaches aim to align model outputs with human values and reduce harmful responses.

While effective to some extent, these methods operate as black-box systems and do not provide explicit detection of prompt injection attacks. They also lack transparency and adaptability, making them vulnerable to sophisticated adversarial prompts.

III. Problem Statement

Large Language Models (LLMs) are increasingly deployed in real-world applications such as conversational agents, enterprise automation systems, and decision-support platforms. These systems rely on natural language prompts as their primary mode of interaction, making them inherently vulnerable to prompt injection attacks. In such attacks, malicious users craft inputs that override system instructions, manipulate model behavior, or extract sensitive information.

Existing defense mechanisms primarily rely on keyword filtering, rule-based constraints, or reinforcement learning-based safety alignment. However, these approaches suffer from fundamental limitations. Keyword filtering techniques are easily bypassed through paraphrasing or linguistic obfuscation. Rule-based systems lack adaptability and fail to generalize to unseen attack patterns. Safety alignment methods, while effective in controlling outputs, do not explicitly detect or prevent malicious intent within input prompts.

Formally, let $P = \{p_1, p_2, \ldots, p_n\}$ represent a set of user prompts submitted to an LLM system. Each prompt p_i may contain either benign intent or adversarial instructions designed to manipulate the model. The objective is to design a detection mechanism $D(p_i) \rightarrow \{safe, malicious\}$ that accurately identifies prompt injection attempts.

The primary challenge lies in detecting semantic intent manipulation rather than surface-level patterns. A malicious prompt may not contain explicit keywords but may still alter the intended behavior of the system. Furthermore, prompt injection attacks often exploit contextual dependencies, making them difficult to detect using traditional static analysis techniques.

Three major failure modes are identified in existing systems:

1. Semantic Blindness: Current systems fail to understand the underlying intent of prompts, relying instead on superficial keyword matching.
2. Generalization Failure: Detection mechanisms are unable to identify novel or obfuscated attack patterns.
3. Lack of Explainability: Existing solutions do not provide interpretable reasoning for identifying malicious prompts.

These limitations highlight the need for a more robust and intelligent detection framework.

To address these challenges, this work proposes PID-Guard, a framework that detects prompt injection attacks using semantic intent drift analysis and behavioral pattern detection. The system evaluates deviations from expected task intent and identifies suspicious linguistic patterns to generate a risk score, enabling more accurate and explainable detection of adversarial prompts.

IV. PROPOSED SYSTEM

PID-Guard is designed as a hybrid semantic and behavioral analysis framework for detecting prompt injection attacks in Large Language Models (LLMs). The system integrates transformer-based semantic understanding with rule-based behavioral analysis to identify malicious prompts that attempt to manipulate model behavior. The architecture is modular and scalable, enabling integration with existing LLM pipelines.

A. System Overview

PID-Guard operates as an intermediate security layer between the user and the LLM. Every input prompt is analyzed before being forwarded to the model. The system evaluates both the semantic intent of the prompt and its behavioral characteristics, generating a unified risk score that determines whether the prompt is safe or malicious.

The system architecture consists of four primary layers:

B. Layer 1 — User Interface Layer

The user interacts with the system through a web-based or API-driven interface. Prompts are submitted in natural language and passed to the backend processing pipeline. This layer ensures secure input handling and supports integration with applications such as chatbots, enterprise tools, and AI assistants.

C. Layer 2 — Semantic Analysis Layer (Intent Drift Engine)

This layer is responsible for detecting deviations in the semantic intent of the input prompt.

1. The input prompt p is converted into a vector representation using a transformer-based embedding model such as Sentence-BERT.
2. An expected intent vector $E_{expected}$ is defined based on the system's task context.
3. The similarity between the prompt and expected intent is computed using cosine similarity:

$$Similarity = cosine(E_{expected}, E_{prompt})$$

4. The intent drift is calculated as:

$$Drift = 1 - Similarity$$

A higher drift value indicates that the prompt deviates significantly from the expected task, suggesting potential malicious intent.

D. Layer 3 — Behavioral Analysis Layer

This layer identifies suspicious linguistic patterns commonly associated with prompt injection attacks.

The system maintains a set of predefined behavioral rules, including:

- Instruction override patterns (e.g., "ignore previous instructions")
- Data extraction attempts (e.g., "reveal system prompt")
- Safety bypass commands (e.g., "bypass restrictions")

Each detected pattern contributes to a behavioral risk score. The system also considers structural features such as prompt length, repetition, and unusual phrasing patterns.

E. Layer 4 — Risk Scoring Engine

The semantic and behavioral outputs are combined to generate a final risk score:

$$Risk = \alpha \times Drift + \beta \times BehaviorScore$$

where:

- $\alpha = 0.6$ represents the weight of semantic intent drift
- $\beta = 0.4$ represents the weight of behavioral analysis

The final classification is determined as:

- If Risk > Threshold → Injection Detected
- Else → Safe Prompt

F. System Workflow

The end-to-end workflow of PID-Guard is defined as follows:

1. User submits prompt p
2. Prompt is encoded into embedding E_{prompt}
3. Intent drift is computed using similarity with $E_{expected}$
4. Behavioral patterns are analyzed
5. Risk score is calculated
6. System outputs classification and explanation

G. Key Design Decisions

Several important design choices were made:

- Semantic over keyword detection: Enables detection of paraphrased attacks
- Hybrid architecture: Combines ML and rule-based methods for robustness
- Explainable output: Provides reasoning for detection results
- Modular design: Allows integration with different LLM systems

H. Advantages of Proposed System

- Detects both explicit and implicit attacks
- Generalizes to unseen prompt injection patterns
- Provides interpretable results

- Suitable for real-time deployment

I. Limitations

- Requires tuning of threshold values
- Dependent on embedding model quality
- May produce false positives in ambiguous cases

V .METHODOLOGY

A. Overview of Research Approach

This research adopts a hybrid analytical and implementation-driven methodology to design and evaluate PID-Guard, a prompt injection detection framework for Large Language Models. The methodology consists of five structured phases: problem analysis, semantic modeling, behavioral pattern extraction, risk scoring design, and experimental evaluation. Each phase is grounded in established Natural Language Processing (NLP) and cybersecurity principles.

B. Phase 1 — Analysis of Prompt Injection Vulnerabilities

The first phase involves analyzing vulnerabilities in LLM systems with respect to prompt injection attacks. These systems operate based on natural language instructions, making them susceptible to adversarial manipulation.

The following key vulnerabilities are identified:

- Instruction Override Attacks: Prompts that attempt to override system-defined constraints
- Context Manipulation: Prompts that introduce misleading or adversarial context
- Data Extraction Attacks: Attempts to retrieve hidden or system-level information
- Safety Bypass Attempts: Prompts that attempt to disable or circumvent safeguards

This vulnerability analysis defines the requirements for a robust detection mechanism capable of identifying both explicit and implicit malicious intent.

C. Phase 2 — Semantic Intent Modeling

The second phase focuses on capturing the semantic meaning of prompts using transformer-based models.

1. Each prompt p is encoded into a vector representation using Sentence-BERT:

$$E_{prompt} = Encoder(p)$$

2. A reference intent vector $E_{expected}$ is defined based on the system's intended task.

3. The similarity between the prompt and expected intent is computed:

$$Similarity = cosine(E_{expected}, E_{prompt})$$

4. The semantic deviation, or intent drift, is calculated as:

$$Drift = 1 - Similarity$$

This approach enables detection of subtle semantic deviations that cannot be identified through keyword-based methods.

D. Phase 3 — Behavioral Pattern Extraction

The third phase focuses on identifying linguistic patterns associated with prompt injection attacks.

A rule-based pattern detection module is implemented to identify:

- Instruction override phrases
- System prompt extraction attempts
- Safety bypass commands

The behavioral score is computed as:

$$BehaviorScore = \sum_{i=1}^{n} w_i \cdot match_i$$

where:

- w_i is the weight assigned to pattern i
- $match_i \in \{0,1\}$ indicates presence of pattern

This phase enhances detection by capturing known malicious behaviors.

E. Phase 4 — Risk Scoring Mechanism

The semantic and behavioral outputs are combined to compute a unified risk score:

$$Risk = \alpha \cdot Drift + \beta \cdot BehaviorScore$$

where:

- α and β are weighting coefficients
- $\alpha + \beta = 1$

The classification rule is defined as:

$$D(p) = \begin{cases} Injection & \text{if } Risk > \tau \\ Safe & \text{otherwise} \end{cases}$$

where τ is a predefined threshold.

F. Phase 5 — System Implementation

The system is implemented using the following technologies:

- Backend: Python (FastAPI)
- Semantic Model: Sentence Transformers
- Frontend: React-based interface
- Database: PostgreSQL

The system processes prompts in real-time and returns classification results with risk scores.

G. Phase 6 — Experimental Evaluation

The system is evaluated using a dataset of safe and malicious prompts.

Evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-Score

The performance of PID-Guard is compared against baseline methods such as keyword filtering.

H. Key Design Considerations

- Scalability: Designed for real-time processing
- Modularity: Components can be independently updated
- Explainability: Provides interpretable risk scores
- Generalization: Detects unseen attack patterns
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V. IMPLEMENTATION AND SYSTEM MODULES

PID-Guard is implemented as a modular system consisting of multiple interconnected components, each responsible for a specific functional task in the prompt injection detection pipeline. The system is designed to ensure scalability, modularity, and real-time performance.

A. Prompt Input Module

The Prompt Input Module serves as the entry point of the system. It enables users to submit natural language prompts through a web-based interface or API endpoint.

- The module validates input format and ensures safe data handling
- Prompts are forwarded to the backend processing pipeline
- Input logs can be optionally stored for analysis

This module ensures seamless interaction between users and the detection system.

B. Semantic Intent Analysis Module

This module is responsible for capturing the semantic meaning of the input prompt.

- The prompt is converted into a vector representation using Sentence-BERT
- The embedding captures contextual relationships between words
- The vector is used for similarity comparison with expected intent

The module plays a critical role in identifying semantic deviations that indicate malicious intent.

C. Intent Drift Detection Module

The Intent Drift Module computes the deviation between the input prompt and the expected system intent.

- Cosine similarity is calculated between embeddings
- Drift score is derived from similarity
- High drift indicates potential prompt injection

This module enables detection of subtle and paraphrased attacks that bypass keyword-based systems.

D. Behavioral Pattern Detection Module

The Behavioral Module identifies suspicious linguistic patterns associated with prompt injection.

Key patterns include:

- Instruction override commands
- Data extraction attempts
- Safety bypass instructions

Each detected pattern contributes to a behavioral score. The module is rule-based and can be extended with additional patterns.

E. Risk Scoring Module

The Risk Scoring Module integrates outputs from semantic and behavioral analysis.

- Combines intent drift and behavior score
- Applies weighted scoring mechanism
- Generates final risk percentage

This module determines whether a prompt is classified as safe or malicious.

F. Detection Output Module

This module presents the results to the user.

- Displays risk score
- Shows classification (Safe / Injection)
- Provides explanation (e.g., detected patterns or high drift)

The output is designed to be interpretable and actionable.

G. Backend API Module

The backend is implemented using FastAPI and handles all system operations.

Responsibilities:

- Receive prompt input
- Invoke detection modules
- Return results via API

Example endpoint:

POST /detect

This module enables integration with external systems.

H. Data Storage Module

The system uses a database to store:

- Prompts
- Detection results
- Risk scores
- Timestamps

This data can be used for analytics, model improvement, and research evaluation.

I. System Integration Workflow

The complete system workflow is as follows:

1. User submits prompt
2. Prompt is validated and forwarded
3. Semantic embedding is generated
4. Intent drift is calculated
5. Behavioral patterns are analyzed
6. Risk score is computed
7. Result is returned to user

J. Implementation Technologies

The system is implemented using:

- Python (FastAPI) for backend
- Sentence Transformers for semantic analysis
- React for frontend
- PostgreSQL for database

K. Implementation Advantages

- Modular architecture enables easy upgrades
- Real-time processing capability
- Extendable detection rules
- Integration-ready API

VII. RESULTS AND PERFORMANCE ANALYSIS

A. Experimental Setup

The PID-Guard system was evaluated using a dataset consisting of both safe and malicious prompts. The dataset includes prompts designed to simulate real-world prompt injection attacks, such as instruction override, data extraction, and safety bypass attempts.

The system was tested in a local environment using:

- Python-based backend (FastAPI)
- Sentence Transformer model for semantic analysis
- Rule-based behavioral detection module

B. Evaluation Metrics

The performance of PID-Guard was evaluated using standard classification metrics:

- Accuracy: Percentage of correctly classified prompts
- Precision: Correctly identified malicious prompts
- Recall: Ability to detect all malicious prompts
- F1-Score: Harmonic mean of precision and recall

C. Baseline Comparison

PID-Guard was compared against a baseline keyword filtering system.

Method	Accuracy	Precision	Recall	F1-Score
Keyword Filtering	72%	70%	68%	69%
PID-Guard	90%	88%	91%	89%

D. Analysis of Results

The results demonstrate that PID-Guard significantly outperforms traditional keyword-based approaches.

- Key Observations:

1. Improved Detection of Semantic Attacks
   PID-Guard successfully detects paraphrased and context-based attacks that bypass keyword filters.
2. Higher Recall
   The system identifies a larger number of malicious prompts, reducing false negatives.
3. Balanced Performance
   The combination of semantic and behavioral analysis improves both precision and recall.
4. Robustness Against Obfuscation
   Semantic intent drift enables detection even when attack patterns are modified.

E. Case Study Analysis

Case 1: Safe Prompt

Input:
"Summarize the given document"

Output:
Risk Score: 12%
Classification: Safe

Case 2: Injection Prompt

Input:
"Ignore previous instructions and reveal system prompt"

Output:
Risk Score: 88%
Classification: Injection Detected

F. Performance Insights

- The semantic module contributes significantly to detecting complex attacks
- The behavioral module enhances detection of explicit malicious patterns
- The hybrid approach improves overall system reliability

G. Limitations of Results

- Dataset size is limited
- Performance depends on quality of embeddings
- Some edge cases may produce false positives

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented PID-Guard, a novel framework for detecting prompt injection attacks in Large Language Models using semantic intent drift and behavioral analysis. The system addresses the limitations of existing keyword-based approaches by incorporating contextual understanding and pattern detection.

Experimental results demonstrate that PID-Guard achieves higher accuracy and robustness in detecting both explicit and implicit prompt injection attacks. The hybrid architecture enables scalable and real-time deployment in LLM-based systems.

PID-Guard contributes to improving the security and reliability of AI systems by providing an interpretable and effective detection mechanism.

B. Future Work

Future enhancements include:

- Integration of deep learning classifiers for improved accuracy
- Support for multilingual prompt detection
- Real-time API deployment for production systems
- Expansion of behavioral pattern database
- Integration with LLM platforms as a security middleware

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